

Evolution of the local environment of lanthanum during simplified SON68 glass leaching

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Abstract

The evolution of the short- and medium-range local environment of lanthanum was determined by L_{III}-edge X-ray absorption spectroscopy (XAS) during leaching of simplified SON68-type glasses. In glass without phosphorus, lanthanum is found in a silicate environment, and its first coordination sphere comprises eight oxygen atoms at a mean distance of 2.51 Å. When this glass was leached at a high renewal rate, the lanthanum local environment was significantly modified: it was present at hydroxycarbonate and silicate sites with a mean La–O distance of 2.56 Å, and the second neighbors consisted of La atoms instead of Si for the glass. Conversely, in the gel formed at low renewal rates, lanthanum was found in a silicate environment similar to that of the glass. In phosphorus-doped glass, lanthanum is found in a phosphate environment, although the Si/P atomic ratio is 20:1. Lanthanum is surrounded by seven oxygen atoms at a mean distance of 2.37 Å. When phosphorus-doped glass is leached, regardless of the leaching solution flow rate, the short- and medium-range lanthanum local environment remains almost constant; the most significant change is a 0.05 Å increase in the La–O distance.

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1. Introduction

France has opted to reprocess spent fuel from its nuclear power plants. This operation selectively recovers the reusable uranium and plutonium separately from the waste fission products and minor actinides. The waste is then immobilized in a glass matrix known as R7T7 glass. Radionuclides are released from the matrix when water comes into contact with the waste package. Leaching of high-level waste containment glass forms an altered layer on the surface of the glass. This layer, or ‘gel’, incorporates a large number of radionuclides, notably transition metals, rare earth elements and actinides. From the perspective of geo-

logical disposal of R7T7 glass, radionuclide retention in the gel significantly diminishes the activity released from the glass package. The persistence of this retention must be demonstrated before it can be taken into account in glass package performance assessment. One way to do so is to study changes in the local environment of the elements during the glass–gel transformation, to determine whether they are in fact incorporated into the gel silicate network. This type of study can yield information concerning the mechanisms of radionuclide incorporation in the gels, as has already been obtained for zirconium [1], iron [1,2], europium [3], cerium [4], neodymium [4], and uranium [5,6].

A significant fraction of the $\beta\gamma$ radioactivity of the R7T7 glass produced at La Hague comes from the rare earth elements, which account for 3.9 oxide wt% of the glass (about one-fourth of the fission products). An L_{III}-edge study of the local environment of cerium and neodymium showed

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that these two rare earth elements exhibit different behavior during glass leaching [4]. We therefore chose to study the local environment of lanthanum during leaching of simplified SON68 glass samples to compare the behavior of lanthanum with that of cerium and neodymium (SON68 is a nonradioactive glass comprising about 30 chemical elements, with the same composition as R7T7 glass). One of the most important glass alteration parameters under geological repository conditions is the groundwater flow rate at the site. Glass samples were altered with different leachate renewal rate, resulting in the formation of gels with different chemical compositions. The influence of phosphorus in the glass on the lanthanum local environment was also studied because of the high affinity of phosphorus for rare earth elements. The lanthanum local environment was investigated by extended X-ray absorption fine structure (EXAFS) which is an ideal structural probe of amorphous materials like glasses and gels.

2. Experimental

2.1. Glass compositions

The test specimens consisted of SON68 glass simplified to 9 or 10 elements, with the same composition as those used to study the local environment of cerium and neodym-

ium. The chemical compositions of the two lanthanum-doped glasses are indicated in Table 1. The glass samples were chemically analyzed by alkaline fusion ($\text{NaOH} + \text{KNO}_3$ and $\text{Li}_2\text{B}_4\text{O}_7 \cdot 5\text{H}_2\text{O}$) of glass powder, then recovered in HNO_3 for ICP-AES analysis. Glass La1, doped with 4 wt% La_2O_3 , was heated for 3 h at 1300 °C in a platinum crucible and annealed for 1 h at 520 °C in a graphite crucible. Glass La2, doped with 4 wt% La_2O_3 and 1.8 wt% P_2O_5 (2 atoms of phosphorus per atom of lanthanum), was heated for 3 h at 1350 °C and annealed for 1 h at 520 °C. Glass La1 was perfectly translucent whereas glass La2 appeared milky and opalescent.

2.2. Gel fabrication

Lanthanum-doped gels were obtained by complete leaching of glasses La1 and La2 under pseudo-dynamic conditions at 90 °C. The gels were prepared from glass powder samples containing the size fraction smaller than 20 μm ; the specific surface area measured by krypton adsorption using the BET method was $0.9 \text{ m}^2 \text{ g}^{-1}$. The uncertainty of 5% associated to this value is due to the error on the Kr adsorption measurement and to the representativeness of the glass powder sample. Like that, the associated uncertainty to SA/V ratio is also 5%. The leaching solution was pure water (18 M Ω cm) and the leachate renewal rates were 2/day, 2/week and 1/week (indicated by the subscripts a, b and c, respectively, in the gel designations). The glass alteration time necessary to obtain gels a, b and c ranged from 1 month to about 1 year. The glass alteration conditions are listed in Table 2. The silicon retention factors were very low in the gels without phosphorus – about 0.1 – but increased to about 0.6 for gels containing phosphorus when the leaching solution renewal rate diminished. The retention factor of an element i $\text{RF}(i)$ is calculated with the following formula: $\text{RF}(i) = 1 - \frac{[i]}{[B]} \times \frac{f_B}{f_i}$ where $[i]$, f_i , $[B]$, f_B are the solution concentration and the fraction in the glass for element i and boron. The uncertainties on B, Si and La solution concentrations are respectively 3%, 3% and 15%. This leads to an uncertainty of ± 0.05 on $\text{RF}(\text{Si})$ and ± 0.01 on $\text{RF}(\text{La})$. Lanthanum was

Table 1
Molar composition of lanthanum-doped glasses and gels

Element	Glasses		Gels				
	La1	La2	La1a	La1b	La2a	La2b	La2c
Si	18.2	17.7	15.9	19.9	16.6	22.1	24.5
Al	1.6	1.6	8.2	5.2	4.0	4.3	3.4
B	9.7	9.8	0.0	0.0	0.0	0.0	0.0
Na	4.2	4.2	1.1	1.9	0.5	0.6	0.4
Ca	1.7	1.6	2.0	1.3	3.7	4.3	3.7
Li	3.2	3.2	1.3	2.9	2.1	0.5	0.3
Zn	0.7	0.7	5.7	2.7	4.0	2.2	1.7
Zr	0.1	0.1	0.8	0.7	0.7	0.3	0.2
La	0.5	0.4	4.2	4.1	3.7	1.3	1.0
P	0.0	0.9	0.0	0.0	6.1	2.7	2.0
O	60.1	59.8	60.8	61.3	58.6	61.7	62.8

Table 2
Characteristics of lanthanum-doped gels

Parameter	La1a	La1b	La2a	La2b	La2c
Glass mass (g)	2.0	2.5	2.0	2.5	1.0
Leachate volume (L)	1.0 ± 0.005	0.5 ± 0.005	1.0 ± 0.005	0.5 ± 0.005	0.06 ± 0.001
SA/V (cm^{-1})	18 ± 0.9	45 ± 2.3	18 ± 0.9	45 ± 2.3	150 ± 7.5
Renewal rate	2/day	2/week	2/day	2/week	1/week
Flow rate/SA ($\pm 5\%$) (m d^{-1})	10^{-3}	5.7×10^{-5}	10^{-3}	5.7×10^{-5}	7.9×10^{-6}
Alteration rate r ($\pm 8\%$) ($\text{g m}^{-2} \text{ d}^{-1}$)	0.042	0.0094	0.140	0.108	0.017
Rate drop r_0/r	24	106	7	9	59
Si retention factor	0.12 ± 0.05	0.10 ± 0.05	0.11 ± 0.05	0.41 ± 0.02	0.60 ± 0.02
La retention factor	0.96 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	1.00 ± 0.01
pH^{90° (± 0.05) start–end	8.0–7.4	7.9–7.4	8.0–7.8	8.8–7.6	8.5–8.0
[Si] ($\pm 3\%$) start–end (mg L^{-1})	11.2–3.1	37.3–13.0	12.2–3.3	90.0–36.5	122–45.6
Gel mass (g)	0.17	0.43	0.42	0.85	0.63

very strongly retained in the gels – even at very high renewal rates – because of its very low solubility. The glass alteration rates ranged from 0.14 to $0.0094 \text{ g m}^{-2} \text{ d}^{-1}$, depending on the leaching solution flow rate. The glass dissolution rate is calculated from the boron normalized mass loss, which is function of SA/V ratio, boron solution concentration and renewal rate. Taking into account the uncertainties on these parameters, the error on the glass dissolution rate is 8%. The phosphorus-doped glass alteration rates were higher than for the gels without P, at the same renewal rate. The alteration rates remained high, compared with the residual long-term alteration rate of SON68 glass, i.e. $1.7 \times 10^{-4} \text{ g m}^{-2} \text{ d}^{-1}$ at 90°C [7]. The gel chemical compositions (Table 1) were calculated from the elemental concentrations analyzed in the leachates because the gel quantities recovered after glass alteration were too small for direct chemical analysis of the solid. Silicon was the major element in the gels, regardless of the leaching solution flow rate. For gels La2b and La2c, the P/La ratio was exactly the same as in the glass, indicating that the behavior of both elements is identical when leached at relatively low flow rates, and that La and P are probably strongly chemically bonded. The lanthanum-doped gel compositions are comparable to those of the neodymium-doped gels [4]; the main difference is sodium and lithium retention in the lanthanum-doped gels, whereas these elements are leached congruently with boron from neodymium-doped gels. Gels obtained after leaching glass powder were dried for 1 day at 90°C to eliminate the free water in the gel pores while conserving the structural water. After drying, the gels were mixed with cellulose, press-compacted and the resulting pellets were wrapped in kapton tape.

It should be noted that the gels have been formed under relatively high flow rates of pure water that do not correspond to the expected conditions in a geological repository. The lowest renewal rate is still about four orders of magnitude higher than the flow rate expected in a disposal site. Like that, the gels were formed in contact with more or less under saturated leaching solutions (different silicon concentrations), which led to depleted gels in very or fairly soluble elements, like calcium or network forming elements (Si, Al). However these leaching conditions had little influence on low soluble elements, like transition metals or rare earth elements. These leaching conditions were chosen to obtain gels in reasonable time, not achievable under representative conditions of geological repository (the glass alteration rate in these conditions is only about 20 nm per year [7]). In a disposal site, the temperature will be about 30°C while the gels were obtained at 90°C . The structure of the gel is independent of the temperature if lower than 100°C [2]. Beyond this temperature, the formation of crystallized phases increases to the detriment of amorphous gel. The goal of this study was to determine the sensitivity of lanthanum local environment to flow rate. This information allows to estimate the lanthanum behavior in repository conditions.

Table 3
Structural parameters of reference compounds

Compound	Oxygen number	La–O (Å)	Space group
La_2O_3	7	2.53	$P\text{-}3m1$
$\text{La}(\text{OH})_3$	9	2.68	$P6_3/m$
$\text{La}_2(\text{CO}_3)_3 \cdot x\text{H}_2\text{O}$	10	2.61	$Pccn$
$\text{LaPO}_4 \cdot x\text{H}_2\text{O}$	7	2.57	$P2_1/c$
LaOHCO_3	10	2.62	$Pmcn$
$\text{La}_2\text{Si}_2\text{O}_7$	8 ^a	2.56 ^a	$P4_1$
LaBSiO_5	10	2.66	$P3_1$
LaAlO_3	12	2.71	$Pm3m$
$\text{Na}_3\text{La}_9\text{O}_3(\text{BO}_3)_8$	8.5 ^b	2.56 ^b	$P\text{-}6_2m$

^a Four sites average.

^b Two sites average.

2.3. XAS reference compounds

Four reference compounds were used for EXAFS data analysis: La_2O_3 [8], $\text{La}(\text{OH})_3$ [9], $\text{La}_2(\text{CO}_3)_3 \cdot x\text{H}_2\text{O}$ [10] and $\text{LaPO}_4 \cdot x\text{H}_2\text{O}$ [11]. Some useful additional reference compounds that were not available – $\text{La}_2\text{Si}_2\text{O}_7$ (La in four crystallographic sites) [12], LaBSiO_5 [13], LaAlO_3 [14], $\text{Na}_3\text{La}_9\text{O}_3(\text{BO}_3)_8$ (La in two crystallographic sites) [15] and LaOHCO_3 [16] – were simulated using the FEFF7 code [17]. The crystallographic data of the reference compounds are listed in Table 3.

2.4. EXAFS data acquisition

The EXAFS spectra were recorded at the LURE synchrotron facility (Orsay, France) on the D44 beam line. This beam line was equipped with a double Si(111) monochromator and borosilicate mirrors in order to remove higher harmonics. Spectra were acquired at the L_{III} -edge of lanthanum ($E_{\text{max}} = 5491 \text{ eV}$) at room temperature in transmission mode using three air-filled ionization chambers. The third chamber was used for energy calibration of the monochromator. The energy calibration was carried out with a vanadium foil ($E_{\text{max}} = 5484 \text{ eV}$). Counts were obtained with current intensities of about 0.3 nA.

2.5. EXAFS data treatment

The EXAFS oscillations were extracted by simulating matrix atomic absorption with a 6th-degree polynomial (EXAFS98 code [18]). The Fourier-transforms (FT) of the EXAFS spectra were calculated using a k^3 -weighted Kaiser window ($\tau = 2.5$) between 2.0 and $9.7 \text{ \AA}^{-1}$. The structural parameters of lanthanum were determined after filtering the signal corresponding to the first (between about 1.4 and $2.3 \text{ \AA}$) and second (between about 2.9 and $3.9 \text{ \AA}$) coordination spheres (distances uncorrected for phase shift) using the RoundMidnight code [18]. Spectra fitted with $k \cdot \chi(k)$ filtering were weighted with $1/s(k)^2$ where $s(k)$ is the standard deviation of the $k \cdot \chi(k)$ spectra calculated using the EXAFSTAT code [18]. The $k \cdot \chi(k)$

spectra were fitted using a value of 0 ± 2 eV for ΔE_0 and a value of 1 for the inelastic electronic loss factor S_0^2 .

When the EXAFS spectra of lanthanum-doped compounds are extracted, occurrence of well known multi-electron excitation [19] (corresponding to the double transition of the $2p \rightarrow 5d^2$ and $4d \rightarrow 5d^2$ electrons for the L_{III} -edge) is observed at about $5.8 \text{ \AA}^{-1}$. This contribution has therefore been subtracted from all EXAFS spectra [20]. The electronic parameters (phase and amplitude, mean free photoelectron range) used for spectra fitting were calculated with FEFF7 from the structure of the reference compounds (Table 3). In absence of reliable estimate of the systematic errors, it is generally accepted that a rather pessimistic estimate of the global errors is $\pm 0.02 \text{ \AA}$ for the nearest-neighbor distances. The uncertainty on the number of first neighbors determined by fitting the EXAFS spectra was about 20%, whereas the uncertainty on the number of second neighbors was very difficult to estimate and largely exceeded 20%.

2.6. Computational procedure

The structures of La1 glass and gel La1b were simulated by classic molecular dynamics (MD) in order to get some La local environments to compare to the EXAFS results using the FEFF7 code. The structure of gel La1a has not been simulated because this compound was highly crystallized. The structure of the phosphorus-doped compounds could not be simulated because the potentials of this element in a silicate glass are not available. The simulated compositions correspond to those in Table 1, but Zn and Zr were disregarded since these elements represent less than 1 at.%, and Li was grouped with Na. The two compounds were prepared by classic molecular dynamics with a time step of 10^{-15} s in the NVE system. Each compound consisted of 5000 atoms. The simulated volumes of glass La1 and gel La1b contained 23 and 225 lanthanum atoms, respectively, each representing a type of site. The first calculation step consisted in equilibrating a liquid state at 4000 K by regularly adjusting the atomic velocities on the setpoint temperature during a 15 ps time interval, followed by quenching at a rate of 10^{14} K s^{-1} to gradually lower the temperature to 293 K. A 7 ps step without temperature control terminated the glass relaxation at room temperature. A final quench at 10^{14} K s^{-1} followed by a relaxation without constraints for 6 ps reduced the glass temperature to near 0 K to immobilize the atom positions around their equilibrium sites. For a given composition, several computations were systematically performed to determine by trial and error the equilibrium volume giving a final structure with virtually zero pressure. The equilibrium densities determined for the two compounds were 2.60 and 2.67 g cm^{-3} , respectively. The potentials used were derived from BMH potentials with additional $1/r^6$ and $1/r^8$ terms to improve the environments of the Na and Ca ions in particular. The details of the formulas and the potential parameter values are reported in references [21,22]. Con-

cerning the potentials involving La ions, the La–O potential was taken from Park et al. [23], and the other Laccation potentials are purely Coulombian. Formal charges were used for each ion. The complete Ewald sum was applied to the Coulombian interactions with a $9 \text{ \AA}$ cutoff radius in real space; the cutoff radius in reciprocal space was fitted using the formulas given by Fincham [24]. For the EXAFS computations around the La ion sites calculated by molecular dynamics, the environment of each lanthanum atom was limited by a $6 \text{ \AA}$ cutoff radius and all the ions inside the cutoff radius were taken into account.

3. Local environment of lanthanum in the glasses and gels

3.1. Glass samples

The EXAFS spectra of glasses La1 and La2 (doped with phosphorus) are compared in Fig. 1. There are two significant differences: a slight shift in k was observed for the fourth oscillation peak near $8.2 \text{ \AA}^{-1}$ for glass La1 and $7.9 \text{ \AA}^{-1}$ for glass La2, together with a small beat near $7.5 \text{ \AA}^{-1}$ for glass La1. Adding even a small quantity of phosphorus thus modifies the nature of the sites occupied by lanthanum in the glass. The Fourier-transforms (FT) of glasses La1 (Fig. 2) and La2 (Fig. 3) indicate that the distribution of La–O bonds (first coordination sphere) is indeed symmetric for both glasses and centered on a single distance. The FTs also suggest the presence of a signal due

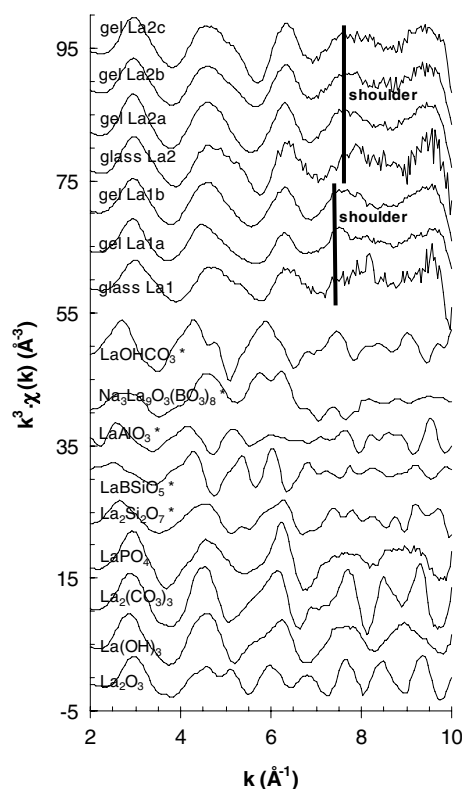


Fig. 1. Lanthanum L_{III} -edge EXAFS spectra of reference compounds, glasses and gels (* = simulated spectrum with FEFF7 code).

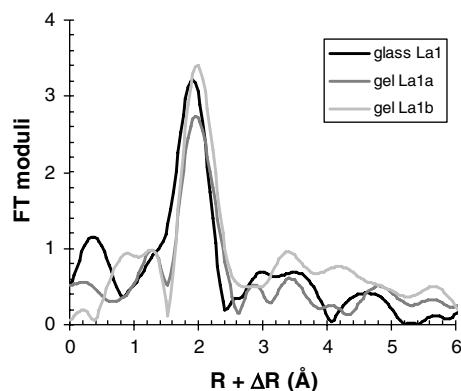


Fig. 2. Radial pseudo-distance distribution in lanthanum-doped glass and gels.

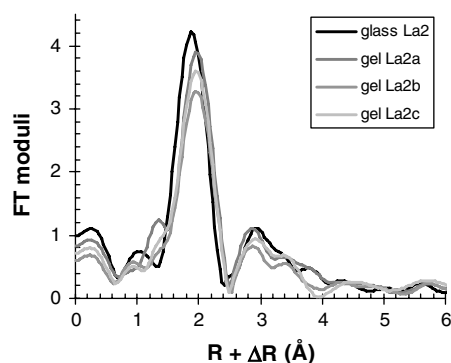


Fig. 3. Radial pseudo-distance distribution in lanthanum- and phosphorus-doped glass and gels.

to the second neighbors at about 3.3 and 2.9 Å (distances uncorrected for phase shift) for respectively glass La1 and La2. Qualitatively speaking, the EXAFS spectrum of glass La1 (Fig. 1), exhibits similarities with lanthanum silicate $\text{La}_2\text{Si}_2\text{O}_7$. This resemblance is not surprising in that silicon is the major element of the glass. The discrepancy between the two spectra could arise from the fact that the $\text{La}_2\text{Si}_2\text{O}_7$ spectrum was calculated from a crystallized compound whereas glass La1 is a fully amorphous material, as shown by its XRD pattern (Fig. 4). The homology of the lanthanum sites in an amorphous material is much lower than in a crystallized material. The EXAFS spectrum of glass La2 resembles that of the reference compound LaPO_4 . Lanthanum thus appears to be present in a phosphate environment in glass La2. The X-ray diffraction pattern of glass La2 (Fig. 4) indicates that this compound is also mainly amorphous (with a characteristic bump between about 15° and 35°) but also includes diffraction lines corresponding to a mixed calcium and lanthanum phosphate $\text{Ca}_8\text{La}_2(\text{PO}_4)_6\text{O}_2$. The presence of this crystallized compound explains the opalescence of glass La2 (compared with glass La1) but SEM–EDS analysis does not show any heterogeneity of the La concentrations in glass La2 – at least at SEM (micrometric) scale.

The experimental $k \cdot \chi(k)$ spectrum of glass La1 (without phosphorus) corresponding to the first coordination sphere

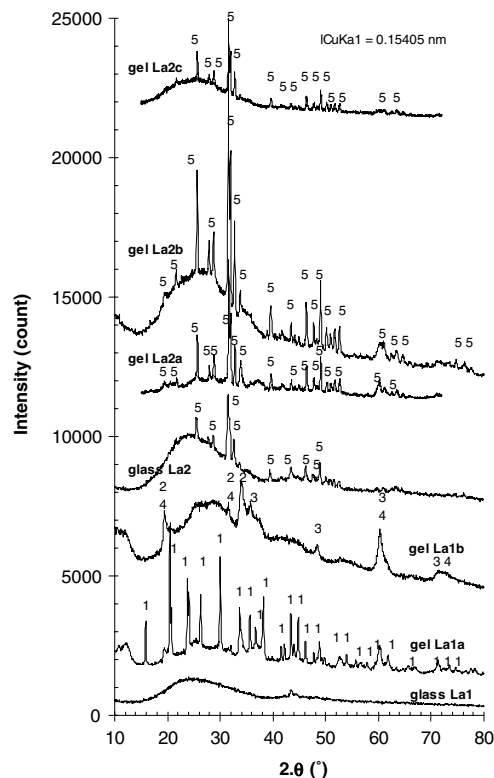


Fig. 4. X-ray diffraction patterns of lanthanum- and phosphorus-doped glasses and gels (1 = LaOHCO_3 ; 2 = $\text{Na}_4\text{Zr}_2(\text{SiO}_4)_3$; 3 = $\text{Li}_2\text{ZnSiO}_4$; 4 = $\text{Na}_{0.3}\text{Zn}_3(\text{Si,Al})_4\text{O}_{10}(\text{OH})_2 \cdot 4\text{H}_2\text{O}$; 5 = $\text{Ca}_8\text{La}_2(\text{PO}_4)_6\text{O}_2$).

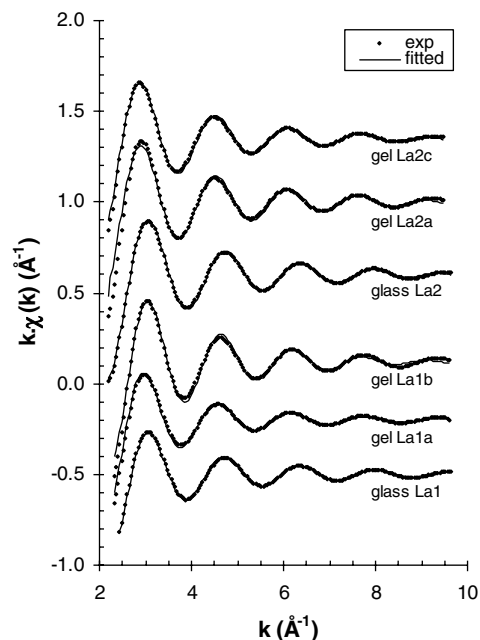


Fig. 5. Comparison between experimental and fitted EXAFS spectra corresponding to the first coordination sphere for lanthanum-doped glass and gels ($\text{La}_2\text{Si}_2\text{O}_7$ site) and lanthanum- and phosphorus-doped glass and gels (LaPO_4 site).

of lanthanum was fitted very satisfactorily (Fig. 5) using the site-averaged phase and amplitude taken from lantha-

Table 4

Structural fitting parameters of EXAFS spectra for the first coordination sphere (La–O) of lanthanum-doped compounds (uncertainty on N is about 20%)

Compound	N	R (Å)	σ (Å)
Glass La1	8.2	2.51 ± 0.02	0.047 ± 0.011
Gel La1a	9.8	2.56 ± 0.02	0.078 ± 0.002
Gel La1b	11.5	2.56 ± 0.02	0.060 ± 0.001
Glass La2	6.7	2.37 ± 0.02	0.096 ± 0.054
Gel La2a	7.3	2.43 ± 0.02	0.091 ± 0.004
Gel La2b	7.5	2.42 ± 0.02	0.084 ± 0.002
Gel La2c	7.9	2.42 ± 0.02	0.084 ± 0.002

num silicate $\text{La}_2\text{Si}_2\text{O}_7$, in which lanthanum occupies four different sites. The EXAFS spectrum of each site was simulated using FEFF7, then the phase and amplitude were extracted from the mean spectrum of the four sites. In glass La1, lanthanum is surrounded by 8.2 ± 1.6 first neighbor oxygen atoms at a mean distance of 2.51 ± 0.02 Å (Table 4). The La–O distance and the number of first neighbors are consistent with the presence of lanthanum in a silicate local environment in glass La1, as the structural data determined for the latter are comparable to those observed for $\text{La}_2\text{Si}_2\text{O}_7$. A study of lanthanum oxide production from lanthanum alkoxides [25] reported La–O distances different from those of La_2O_3 . The fact that the La–O distances in glass La1 differ from those of the reference compound $\text{La}_2\text{Si}_2\text{O}_7$ is thus not incompatible with the presence of lanthanum in a silicate environment in glass La1. A study of La_2O_3 -doped ternary $\text{SiO}_2\text{--Na}_2\text{O--Al}_2\text{O}_3$ peralkaline glass (9.18 wt% $\text{Al}_2\text{O}_3 \cdot \text{Na}_2\text{O}$) [26] showed that lanthanum was surrounded by seven oxygen atoms at a mean distance of 2.53 Å. In the case of an albite composition (19.12 wt% $\text{Al}_2\text{O}_3 \cdot \text{Na}_2\text{O}$), the lanthanum environment changes significantly: it comprises 8–9 oxygen atoms at a mean distance of 2.59 Å. A molecular dynamics study of ternary $\text{La}_2\text{O}_3\text{--Na}_2\text{O--SiO}_2$ glass [23] reported a mean coordination of 6.5 oxygen atoms at a mean distance of 2.50 Å. The La–O distance of glass La1 is very close to that of the peralkaline glass, which has practically the same simplified composition as glass La1. Conversely, the lanthanum coordination number appeared to increase slightly with the number of oxides present in the glass. Fitting the EXAFS spectrum on the second coordination sphere of glass La1 (Fig. 6) resulted in good agreement between the fitted and experimental spectra by assuming that the lanthanum second neighbors consisted of silicon atoms. The fit was based on the La–O–Si phase and amplitude calculated from the reference compound $\text{La}_2\text{Si}_2\text{O}_7$. The second lanthanum coordination sphere in glass La1 thus consists of about three silicon atoms at a mean distance of 3.84 ± 0.02 Å (Table 5). A study carried out on a seven-oxide glass doped with up to 16 wt% Nd_2O_3 [27] showed that the second neodymium coordination sphere comprised 3.5 silicon atoms at a mean distance of 3.91 Å, very near the values found for lanthanum in glass La1. The presence of silicon atoms as second neighbors of lanthanum corrob-

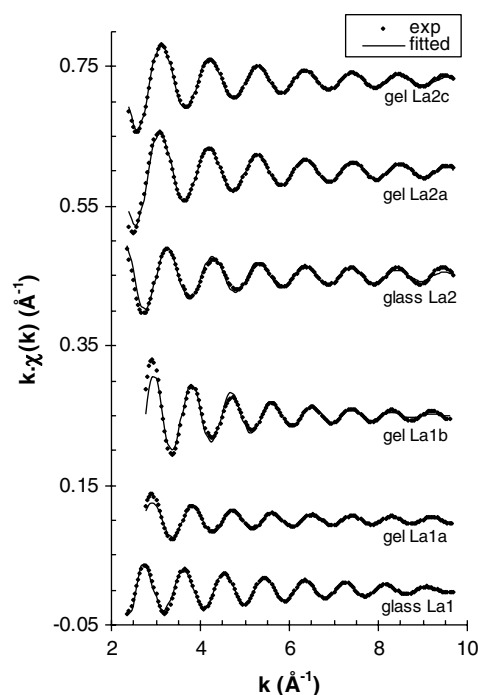


Fig. 6. Comparison between experimental and fitted EXAFS spectra corresponding to the second coordination sphere for lanthanum-doped glass and gels (La–O–Si except for gel La1a (La–O–La)) and lanthanum- and phosphorus-doped glass and gels (La–O–P).

Table 5

Structural fitting parameters of EXAFS spectra for the second coordination sphere of lanthanum-doped compounds (uncertainty on N is expected higher than 20%)

Compound	Second neighbors	N	R (Å)	σ (Å)
Glass La1	Si	2.8	3.84 ± 0.02	0.102 ± 0.029
Gel La1a	La	6.0	3.56 ± 0.02	0.114 ± 0.014
Gel La1b	Si	6.0	3.78 ± 0.02	0.155 ± 0.015
Glass La2	P	1.0	3.23 ± 0.02	0.022 ± 0.015
Gel La2a	P	1.1	3.19 ± 0.02	0.042 ± 0.015
Gel La2b	P	0.8	3.18 ± 0.02	0.015 ± 0.007
Gel La2c	P	1.1	3.20 ± 0.02	0.038 ± 0.013

orates the presence of lanthanum in a silicate environment in glass La1.

As the EXAFS spectrum of glass La2 (with phosphorus) strongly resembles the reference compound LaPO_4 , the $k \cdot \chi(k)$ experimental spectrum of glass La2 corresponding to the first lanthanum coordination sphere was fitted by assuming a phosphate site. The phase and amplitude were extracted from the LaPO_4 EXAFS spectrum calculated with FEFF7, resulting in a very satisfactory fit for the spectrum of glass La2 (Fig. 5). The first lanthanum coordination sphere in glass La2 consists of 6.7 ± 1.3 oxygen atoms at a mean distance of 2.37 ± 0.02 Å (Table 4). A satisfactory fit of the second coordination sphere of glass La2 (Fig. 6) was obtained using the La–O–P phase and amplitude calculated from the LaPO_4 reference compound. The fit results in about 1 phosphorus atom at a mean distance of 3.23 ± 0.02 Å from the lanthanum atoms (Table 5). The La–O distance found in glass La2 is considerably

shorter than for glass La1 ($-0.14 \text{ \AA}$) and for LaPO_4 ($-0.20 \text{ \AA}$), whereas the La–P distance of glass La2 is identical with that of the LaPO_4 reference compound ($3.24 \text{ \AA}$). The La coordination number in glass La2 is also slightly lower than in glass La1, although the variations in the number of neighbors around lanthanum are within the experimental uncertainty range. The differences between the ionic radii ($\text{La}^{3+}(\text{VIII})$ – $\text{La}^{3+}(\text{VII})$) and ($\text{La}^{3+}(\text{VIII})$ – $\text{La}^{3+}(\text{VI})$) are $0.06 \text{ \AA}$ and $0.13 \text{ \AA}$, respectively [28]. Given the uncertainty on the number of first neighbors, the difference in the La–O distance between the two glasses could be attributable simply to the difference in ionic radii due to the different lanthanum coordination numbers of the two compounds. It was impossible to fit the experimental spectra of glass La2 (first and second neighbors) with the phases and amplitudes of $\text{La}_2\text{Si}_2\text{O}_7$ and La–O–Si, however, confirming that La is primarily found in a phosphate environment in this glass and thus very different from that of glass La1. A study of $(\text{Ln}_2\text{O}_3)_x(\text{P}_2\text{O}_5)_{1-x}$ glasses [29] showed that the coordination number of the rare earth elements diminished from 8 to 6 between the ultraphosphates ($x < 0.15$) and metaphosphates ($x \sim 0.25$). Another study [30] on the same type of material showed that the lanthanum coordination number varied from 8.2 ($x \approx 0.13$) to 6.2 ($x \approx 0.25$) for a La–O distance of $2.46 \text{ \AA}$. For a lanthanum phosphate glass corresponding to $x = 0.2$ [31] La is surrounded by 5.9 oxygen atoms (1st layer) and 3.6 atoms phosphorus (2nd layer) at $2.41 \text{ \AA}$ and $3.61 \text{ \AA}$, respectively. The La2 glass composition gives $x \sim 0.5$: La should therefore be present in a metaphosphate environment. This is indeed the case, as the coordination number and La–O distance in glass La2 largely correspond to this environment. Lanthanum thus be found in a phosphate environment with La–O distances very different from those encountered in LaPO_4 . Lanthanum is present in a phosphate environment in glass La2 even though there are 20 silicon atoms per phosphorus atom in this compound; this is consistent with the very high chemical affinity of the lanthanides for phosphates [32].

3.2. Gel samples

The EXAFS spectra of gels La1ab (without phosphorus) are relatively similar to those of glass La1 (Fig. 1) except for the strong beat at $7.5 \text{ \AA}^{-1}$, which is particularly visible for gel La1a and reflects the influence of the leachate renewal rate. Conversely, the three (phosphorus-doped) gels La2abc are practically identical with one another (Fig. 1), indicating the minor effect of the leaching solution flow rate on the lanthanum local structure, and exhibit little difference with glass La2 other than the appearance in the gels of a beat near $7.7 \text{ \AA}^{-1}$. The FTs for gels La1ab (Fig. 2) show no differences for the first lanthanum coordination sphere in the two materials. Conversely, the signal arising from the second coordination sphere indicates that the second neighbors of La are situated at a greater distance in gel La1b than in gel La1a. The FTs of gels La2abc

are identical with one another and with that of glass La2 (Fig. 3), indicating that the leaching solution flow rate has little effect on the short- and medium-range lanthanum local environment when the glass contains phosphorus. The EXAFS spectrum of gel La1a exhibits many similarities with that of the hydroxycarbonate and, to a lesser extent, of the silicate, whereas there are no real correlations between the spectrum of gel La1b and those of the reference compounds. The characteristic bump of amorphous glass is almost absent from the X-ray diffraction pattern of gel La1a (Fig. 4) and all the diffraction lines correspond to lanthanum hydroxycarbonate LaOHCO_3 . Conversely, the bump between 15° and 35° is still visible for gel La1b, with a few broad peaks corresponding to poorly crystallized compounds attributed to $\text{Na}_4\text{Zr}_2(\text{SiO}_4)_3$, $\text{Li}_2\text{ZnSiO}_4$ and $\text{Na}_{0.3}\text{Zn}_3(\text{Si},\text{Al})_4\text{O}_{10}(\text{OH})_2 \cdot 4\text{H}_2\text{O}$. Lanthanum is thus not a constituent of the crystallized phases and is probably incorporated in the amorphous silicate network of gel La1b. The EXAFS spectra of gels La2abc strongly resemble that of the reference compound LaPO_4 (Fig. 1) and the XRD patterns (Fig. 4) of the three gels are identical. The diffraction lines correspond to the same compound identified in glass La2, i.e. a mixed calcium and lanthanum phosphate $\text{Ca}_8\text{La}_2(\text{PO}_4)_6\text{O}_2$. Lanthanum is thus probably present in a phosphate environment in gels La2abc.

The $k \cdot \chi(k)$ EXAFS spectra corresponding to the first coordination sphere of gels La1ab were fitted using the same phase and amplitude as for glass La1. The resulting fit was very satisfactory for both gels (Fig. 5). In gel La1a, lanthanum is surrounded by 9.8 ± 2.0 oxygen atoms at a mean distance of $2.56 \pm 0.02 \text{ \AA}$ (Table 4) whereas in gel La1b the fit indicates 11.5 ± 2.3 oxygen atoms also at a distance of $2.56 \pm 0.02 \text{ \AA}$. Although the variations in the coordination number between gels and glass are within the EXAFS measurement uncertainty range, considering that all the spectra were processed and fitted in the same way, the trend toward a higher coordination number in the gels appears to be significant. The glass transformation into a gel is also accompanied by a slight increase in the La–O distance ($+0.05 \text{ \AA}$). Given the uncertainty on the number of neighbors, the difference in this distance could be attributable to the increase in the lanthanum coordination number from VIII to IX [28]. Fitting the signal for the second lanthanum coordination sphere in gel La1a yielded about 6 lanthanum atoms at a mean distance of $3.56 \pm 0.02 \text{ \AA}$, whereas in gel La1b the second layer consists of about six silicon atoms at a mean distance of $3.78 \pm 0.02 \text{ \AA}$ (Table 5). The leaching solution renewal rate – and thus the mean rate at which the glass is transformed into gel – thus has a major effect on the medium-range lanthanum local environment, reflecting the different mechanisms of lanthanum incorporation in the gels for these two materials. In gel La1a the second neighbors consist of lanthanum atoms, indicating that for a high glass alteration rate La is incorporated in the gel by a mechanism of dissolution and precipitation of lanthanum-rich phases. Conversely, in gel La1b, corresponding to a much lower

glass alteration rate, the second neighbors of La consist of silicon atoms, as in the glass. The distance La–Si in gel La1b is near that of the glass (-0.06 Å) and a significant increase in the number of Si atoms is observed compared with the glass. During the formation of gel La1b, the absence of any La–La contribution indicates that the structure around the lanthanum atom is practically unchanged (compared with the glass) and its new local environment results simply from the release of mobile glass constituents (B, Na, Li) leading to an in situ rearrangement around the lanthanum atom, since the latter does not participate in the crystallized phases present in gel La1b. Lanthanum hydroxycarbonate LaOHCO_3 was identified in gel La1a, but the La–O and La–La distances measured by EXAFS do not correspond to this compound. In addition, the phase and amplitude extracted from the LaOHCO_3 reference compound result in a very satisfactory fit of the spectrum for gel La1a, but the structure does not correspond to LaOHCO_3 (distance and number of first neighbors). One possibility is that lanthanum is not incorporated in gel La1a only as LaOHCO_3 , but occupies at least two different sites, one of which silicated, as has already been observed in comparable materials doped with Eu [3], Ce, or Nd [4]. This assumption is in agreement with the higher value of Debye-Waller factor for gel La1a. The EXAFS spectrum of gel La1a is quite similar to that of $\text{La}_2\text{Si}_2\text{O}_7$ (Fig. 1), which would seem to indicate that lanthanum is also in a silicate environment in this compound. A study of the lanthanum local environment in the interlayer space of moist montmorillonite [33] showed that La was surrounded by nine oxygen atoms at 2.58 Å and that its coordination number diminished when the clay was dried at 110°C . It is thus likely that lanthanum does not occupy this type of site in gels La1ab, but is incorporated in the gel-forming silicate network (gel La1b) or forms lanthanum-rich precipitates in the gel (gel La1a). Lanthanum incorporated in materials such as allanite $(\text{Ca,REE})_2(\text{Al,Fe})_3(\text{SiO}_4)_3\text{OH}$ is found at coordination number 11 [34] and could thus be a possible model for sites occupied by La in gel La1b. Examining the influence of the leaching solution renewal rate on the local environment of the rare earth elements as the glass is transformed into gel (for materials not doped with phosphorus) showed that the behavior of lanthanum resembles rather that of Ce^{4+} more than Nd^{3+} , which is relatively surprising.

The EXAFS spectra of phosphorus-doped gels were fitted with same phases and amplitudes (LaPO_4 and La–O–P) as those used for the spectra of glass La2. Very satisfactory fits were obtained for both the first (Fig. 5) and second (Fig. 6) coordination spheres. The fitted experimental spectra result correspond to an identical short- and medium-range local environment for the three gels La2abc: the first sphere consists of about 7.6 ± 1.5 oxygen atoms at 2.42 ± 0.02 Å on average (Table 4) and the second sphere of about 1.0 ± 0.5 phosphorus atom at 3.19 ± 0.02 Å on average (Table 5). As for the gels containing no phosphorus, an increase of 0.05 Å in the La–O distance was

observed in gels La2abc compared with glass La2. This increase could be attributable to a lanthanum coordination number one unit higher in gels La2abc than in the glass, although the uncertainty on the number of neighbors prevents us from validating this hypothesis. At the most, there appears to be a slight trend toward higher lanthanum coordination numbers during La2 glass alteration. The glass and the gels have the same number of second neighbor phosphorus atoms, which is theoretically surprising – particularly compared with gel La1b in which there was a significant increase in the number of silicon atoms compared with glass La1. For gels La2abc the invariance in the number of second neighbors compared with the glass could be accounted for by assuming the lanthanum phosphate sites are not connected to the gel aluminosilicate network but are dispersed in it. This could also account for the fact that the leaching solution flow rate does not affect the lanthanum local environment. That the short- and medium-range local environment remains practically unchanged as the phosphorus-doped glass is transformed into a gel indicates that the phosphate sites are highly stable under leaching. This stability is confirmed by the fact that the lanthanum local environment is not sensitive to the glass alteration rate, since all three gels La2abc have the same structural parameters. When the glass is doped with phosphorus, lanthanum exhibits the same behavior as neodymium during glass alteration (unaffected by the leachate renewal rate).

4. Discussion

The fitted experimental spectra for the compounds not doped with phosphorus indicate a silicate environment with differences in La–O and La–Si distances compared with the $\text{La}_2\text{Si}_2\text{O}_7$ reference compound.

The EXAFS spectrum of each La atom was simulated with FEFF7 based on the lanthanum sites calculated by molecular dynamics. The mean EXAFS spectra of the simulated glass La1 and gel La1b obtained by averaging each contribution of each La atom were compared with the experimental spectra of these two materials. The short- and medium-range local environment of lanthanum calculated by molecular dynamics (Table 6) does not show any differences between the two simulated compounds because the computation was based on the chemical compositions of the glass and gel, which are not markedly different. For glass La1 the calculation gives a coordination number for first neighbor oxygen atoms of about 6.5, rather than the experimental value of 8. Conversely, the calculated La–O distance (2.52 Å) is practically identical to the value determined by EXAFS (2.51 Å). For the second lanthanum coordination sphere, the La–Si distance is also in good agreement, although the calculated number of silicon atoms is higher than that determined by EXAFS. Conversely, there is a significant discrepancy between the calculated and experimental $k^3 \cdot \chi(k)$ EXAFS spectra (Fig. 7), reflecting major differences between the structure calculated by MD and the structure of glass La1. This difference

Table 6
Average structural parameters of glass La1 and gel La1b calculated by molecular dynamic

Compound	First neighbors	N	R (Å)	Second neighbors	N	R (Å)
Glass La1	O	6.5	2.52	Si	5.8	3.82
				B	3.3	3.50
				Al	<0.1	3.74
				La	<0.1	4.30
Gel La1b	O	5.6	2.54	Si	5.7	3.82
				Al	1.2	3.78
				La	1.7	4.05

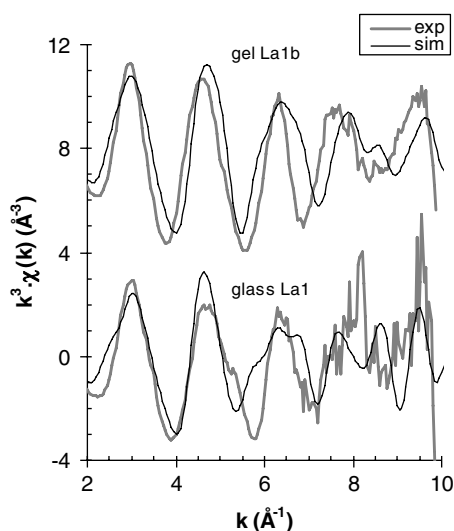


Fig. 7. Comparison between experimental and simulated (from structure calculated by MD) EXAFS spectra.

is due first of all to the very small number of lanthanum atoms (only 23) present in the glass simulated by MD, whose local environment is also very different from one lanthanum atom to the another. It would also probably be necessary to refine the lanthanum potentials in the MD simulation, as the values used were determined for a ternary $\text{SiO}_2\text{--Na}_2\text{O--La}_2\text{O}_3$ glass whereas the simulated La1 glass was much more complex (containing 6 oxides). A comparison between the experimental and simulated EXAFS spectra for gel La1b reveals a shift in the oscillations due to a difference in the interatomic distances, even though the overall appearance of the simulated spectrum is close to that of the experimental spectrum (Fig. 7). The mean La–O distance determined by MD is about 0.02 Å shorter, whereas the mean calculated La–Si distance is about 0.04 Å longer (Table 6). It should be noted that the formation conditions of gel La1b by MD are not physically representative (the quenching by MD is different from leaching) and can explain the differences in structure, or at least in part, between MD and EXAFS. Considering the differences in the local environment of the lanthanum atoms obtained when simulating La-doped structure materials by MD, the volume of simulated material must include at least 150 lanthanum atoms for the calculated

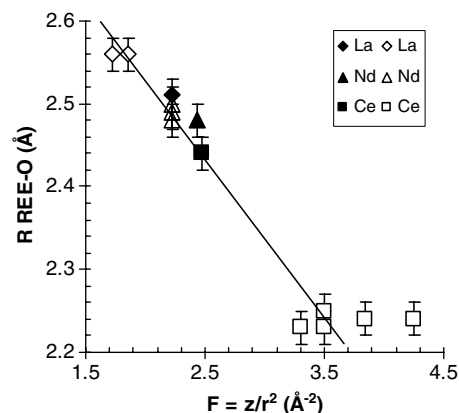


Fig. 8. REE–oxygen interatomic distances versus field strength of REE (full and hollow symbols correspond respectively to glasses and gels). A line is drawn to indicate the tendency of the evolution of distances versus F .

EXAFS spectrum to be independent of the number of La atoms. Finally, these MD calculations showed that it is possible to have lanthanum in a silicate environment with distances different from those of $\text{La}_2\text{Si}_2\text{O}_7$. This is backed by our EXAFS results.

Studies of changes in the local environment of lanthanum, cerium and neodymium during glass alteration have shown that La behavior (coordination number and chemical nature second neighbors) was closer to that of Ce than of Nd [4] for materials not doped with phosphorus. Conversely, with phosphorus-doped materials, La exhibits the same behavior as Nd and probably Ce, although the study was not carried out on materials doped with both Ce and P. The REE–O distances in the glass varied relatively little with the nature of the REE: the longest distances were obtained for lanthanum and the shortest for cerium, whereas in the gels the Ce–O distances were much shorter than the La–O and Nd–O distances, probably because of the change in the oxidation state of cerium as the glass was transformed into a gel. The field strength F of the REE ions is defined as follows: $F = z/r^2$ where z represents the electric charge of the ion and r the ionic radius of the cation. Plotting the REE–O distances determined by EXAFS for La, Ce and Nd versus the field strength yields a linear relation, as shown in Fig. 8, despite two outlying points corresponding to Ce-doped gels obtained at low renewal rates. The changing field strength could account for the variations in the REE–O distances between different rare earth elements, and between the glasses and gels. The greater the rare earth field strength, the more strongly it attracts the O^{2-} ion, resulting in a shorter REE–O distance.

5. Conclusion

The evolution of the short- and medium-range local environment of lanthanum (glass → gel transformation) was determined during leaching of a simplified SON68-type glass.

In the glass without phosphorus, lanthanum is found incorporated at silicate sites. As this glass is altered, the lanthanum local environment undergoes major changes depending on the glass leaching solution flow rate. In the gel obtained at a high renewal rate (2 per day) the lanthanum coordination number and La–O distance increased compared with the glass; La is probably present at two different types of sites: LaOHCO_3 and a silicate. The lanthanum environment is largely reorganized in this gel, with all the second neighbors being lanthanum atoms in place of the silicon atoms in the glass. In the gel formed at a lower renewal rate (2 per week) the lanthanum coordination number and La–O distance are comparable to the gel formed at a high alteration rate but the structural changes are minor compared with the glass because lanthanum remains at silicate sites (silicon second neighbors). This study has shown that – at least for compounds not containing phosphorus – the lanthanum behavior during glass leaching resembles that of cerium more than neodymium.

In phosphorus-doped glass, lanthanum is found in a phosphate environment with a lower coordination number and shorter La–O distance than in a silicate environment. When this glass is leached, the lanthanum local environment is practically unaffected; the main difference between the glass and gels is a small increase in the La–O distance. Moreover, the short- and medium-range lanthanum local environment does not vary when the leaching solution renewal rate is reduced from twice a day to once a week, reflecting the high alteration stability of phosphate sites. Although the Si/P atomic ratio is 20:1 in the phosphorus-doped glass, lanthanum is present in a phosphate environment, confirming the very high affinity of lanthanum for phosphorus.

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